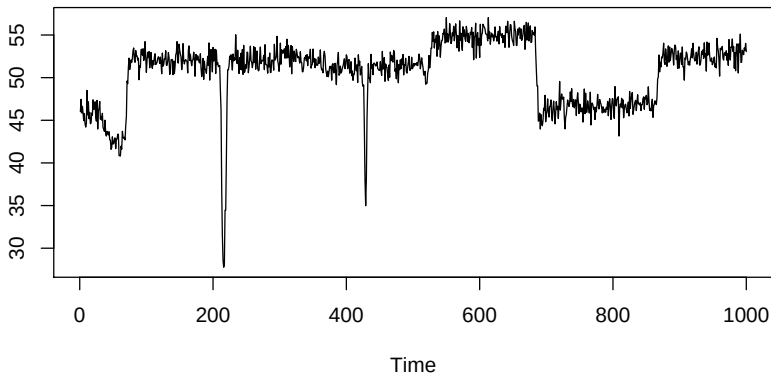


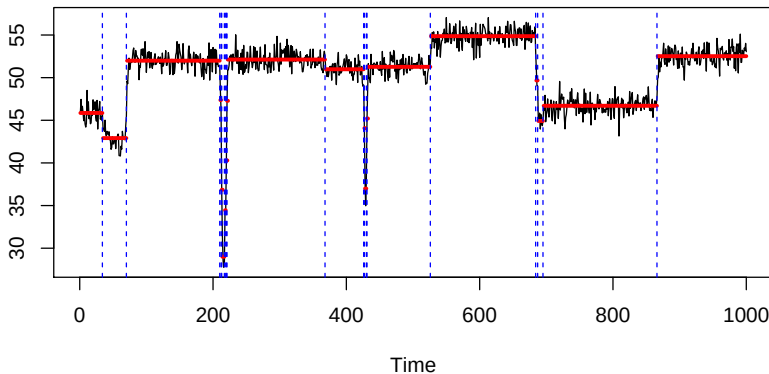
Influence in the world of changepoints

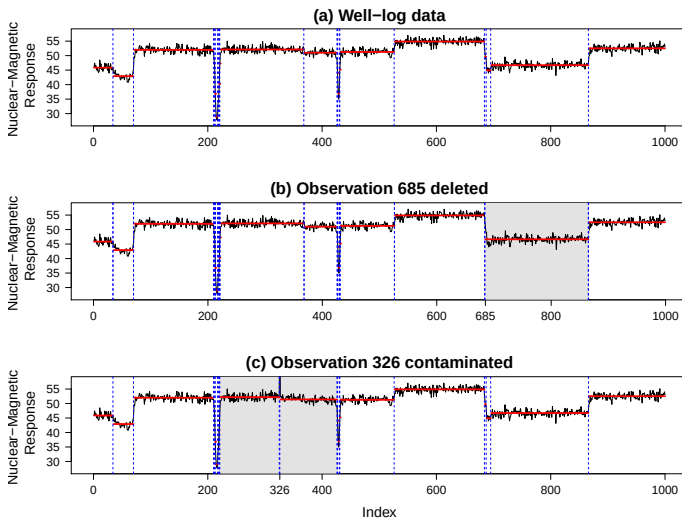
Rebecca Killick(r.killick@lancs.ac.uk)
ICORS July 2022



- Motivation
- Influence for changepoints
- Assessing Influence
- Applications







- Which data points are **influential** for obtaining the segmentation?
 - Changepoints versus Outliers
 - How to measure influence?
- How **stable** is the obtained segmentation?



Sources of Inspiration:

- Regression Analysis: Measures of Influence (e.g., Cook's distance)
- Robust Statistics: Influence Functions

Two routes:

- Modifying an observation
- Leaving out an observation



Roll through all observations $i = 1, \dots, n$ and repeat:

- **Modify** observation i (Make an outlier)
- Obtain the **Expected Outlier Segmentation** (from theory)
- Obtain the **Actual Outlier Segmentation** (from preferred changepoint algorithm)

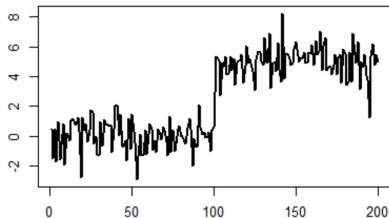
Outcomes:

- Changepoint remains in all segmentations
- Changepoint is deleted in some segmentations
- A new location is a changepoint in some segmentations

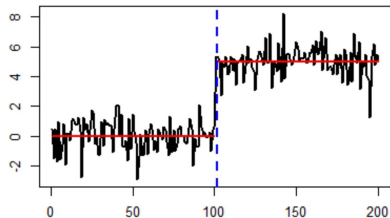
Assessing Influence: Modify



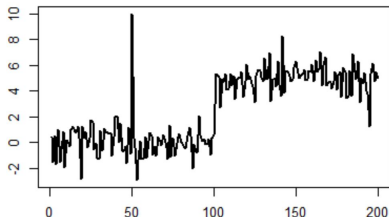
Change in Mean



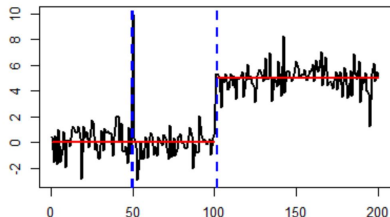
Segmentation



Change in Mean with Outlier

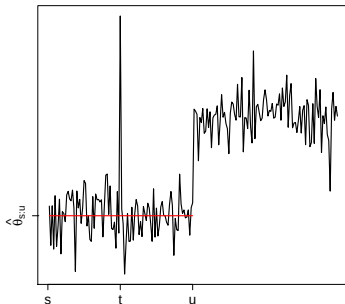


Segmentation with Outlier





Segmentation with Outlier



Let

- y_t be an **outlier**
- $y_{s:u}$ be segment containing the outlier.

Two extra changepoints at $t - 1$ and t lead to a reduction in cost if

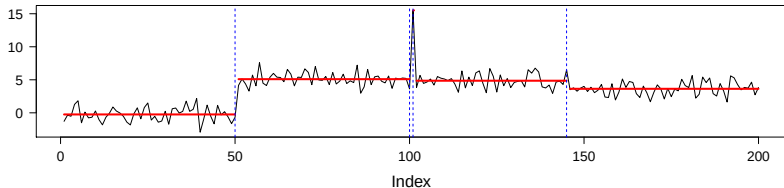
$$2\beta < \gamma(y_t, \hat{\theta}_{s:u})$$

where $\hat{\theta}_{s:u} = \operatorname{argmin}_{\theta} \sum_{j=s}^u \gamma(y_j; \theta)$ and $\gamma(\cdot)$ is the cost.



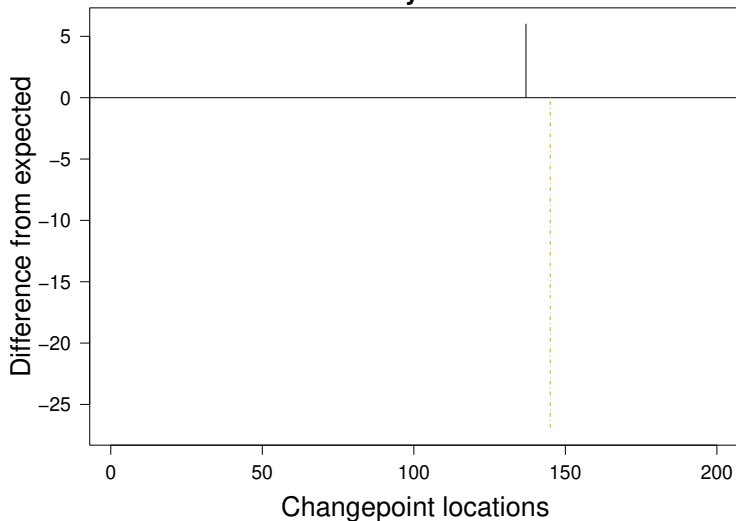
Running example:

- One large change
- One outlier
- One small change



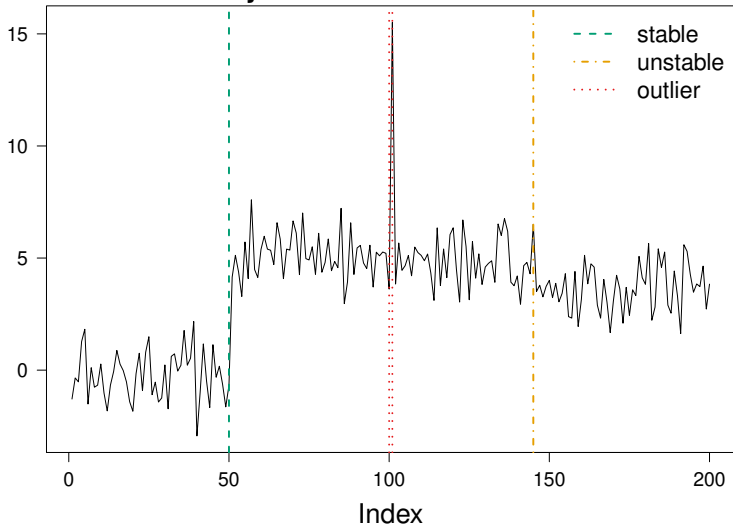


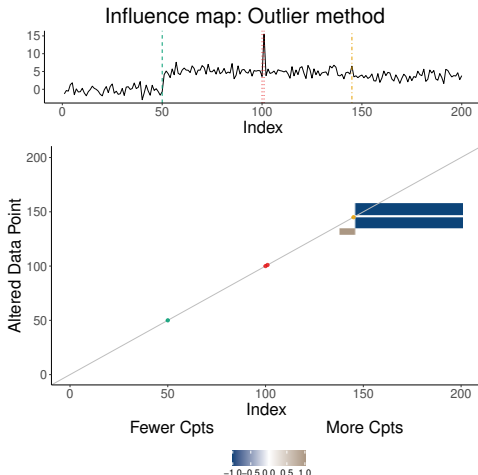
Location Stability: Outlier method



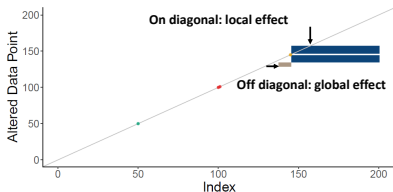
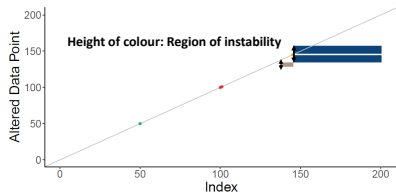
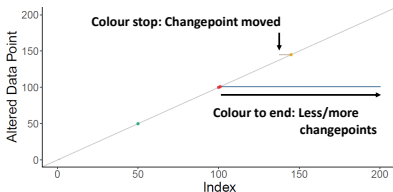
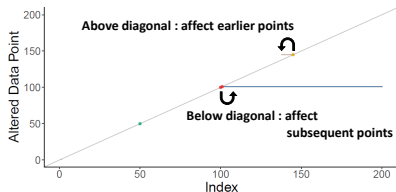


Stability Dashboard: Outlier method



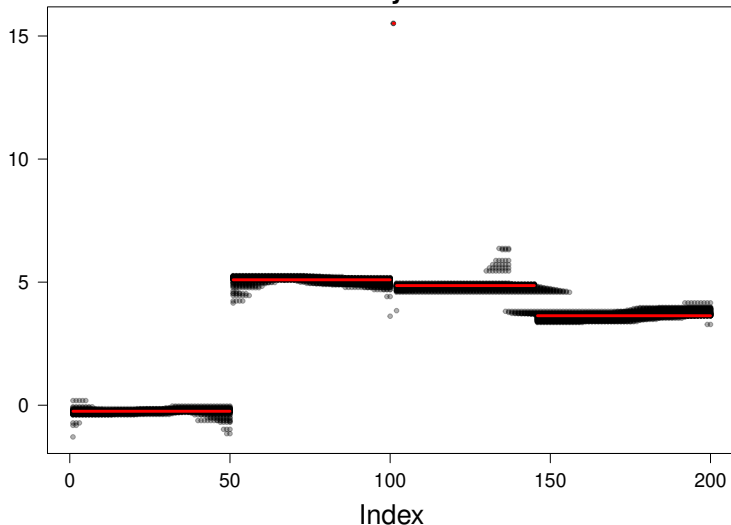


Assessing Influence: Map

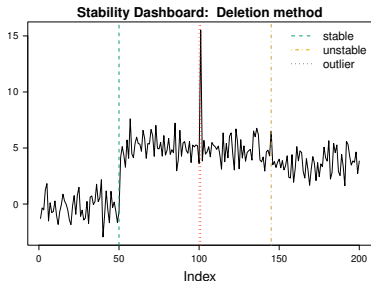
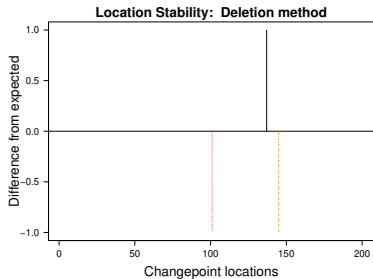




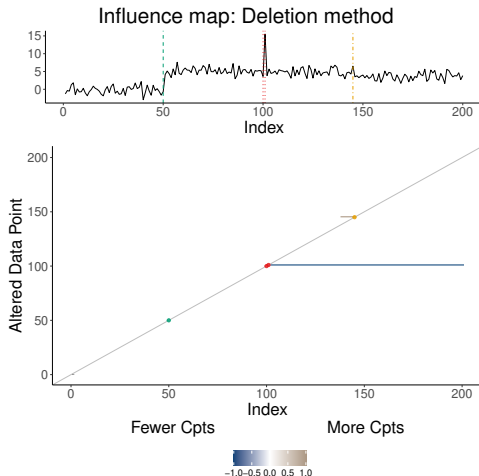
Parameter Stability: Outlier method



Assessing Influence: Deletion



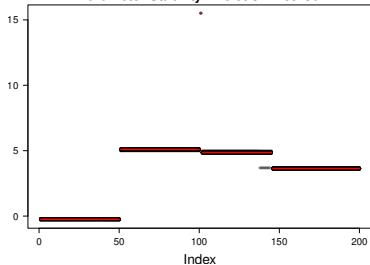
Assessing Influence: Deletion



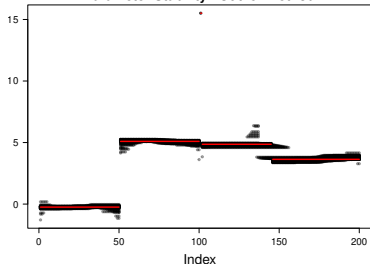
Assessing Influence: Deletion



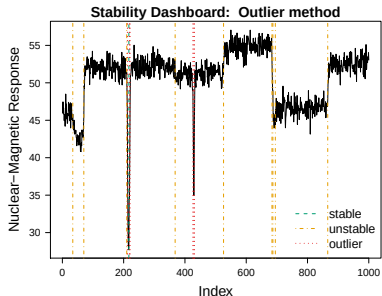
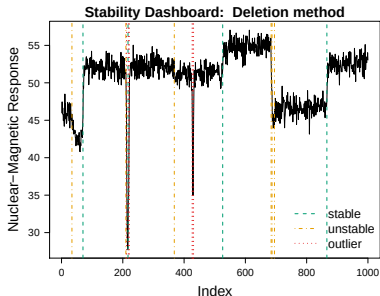
Parameter Stability: Deletion method



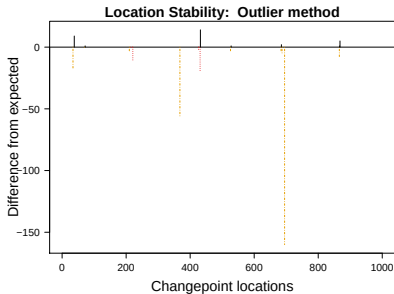
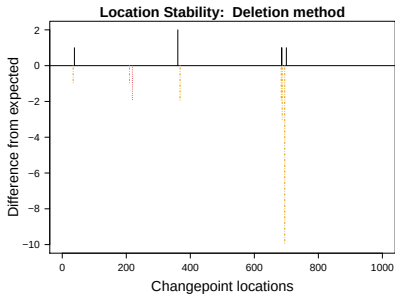
Parameter Stability: Outlier method



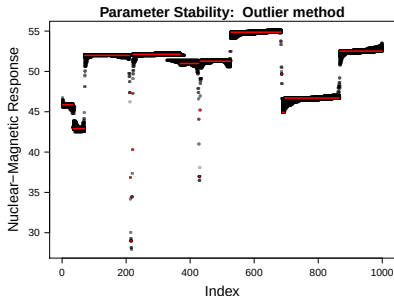
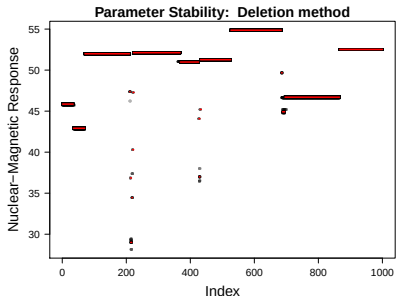
Application: Well-log



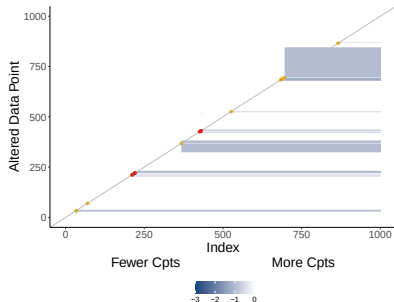
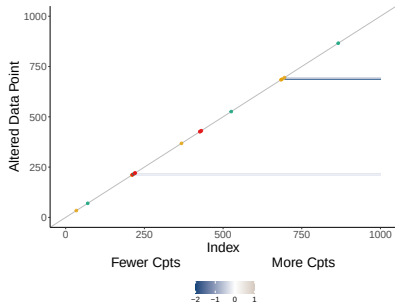
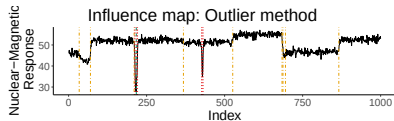
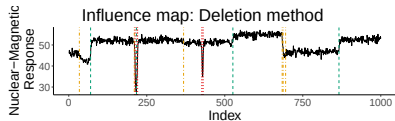
Application: Well-log



Application: Well-log



Application: Well-log



- Indications of stability of changepoints is important for practitioners
- Decisions based on unstable changepoints may be flawed
- Having informative plots for assessing stability is important
- Modifying or deleting individual data points can give an indication of stability
- There are plenty more measures of stability we could use!

Wilms, Killick, Matteson (2021+) Graphical Influence Diagnostics for Changepoint Models. JCGS

www.tandfonline.com/doi/full/10.1080/10618600.2021.2000873